Cristian Enrique Muñoz Villalobos

Rio de Janeiro, Brazil crisstrink@gmail.com 55 21 997262400 crismunoz.github.io
crismunozv crismunoz

Education

Pontifical Catholic University of Rio de Janeiro (PUC-Rio) August 2015 to August 2019
Ph.D. in Electrical Engineering

Pontifical Catholic University of Rio de Janeiro (PUC-Rio) March 2012 to March 2014
M.Sc. in Electrical Engineering

National University of Engineering (UNI) June 2005 to July 2010
B.Sc. in Mechatronics Engineering

Experience

Senior Research - Machine Learning Engineer *London, UK*
Holistic AI *May 2022 – present*

- **Developed benchmarks** for evaluating Large Language Models (LLMs) across key technical risks, including efficacy, robustness, and privacy, deploying these solutions on **Azure Platform** and **Google Cloud Platform (GCP)**.
- **Engineered a risk mitigation tool** for AI systems, addressing issues related to model safety, security, and performance, using **Jax**/**Optax**/**Flax**.
- **Led the development of an AI trustworthiness toolkit**, defining Responsible AI metrics and integrating advanced bias mitigation techniques for bias, fairness, and explainability.

Data Science Advisor *Rio de Janeiro, Brazil*
Applied Computational Intelligence Laboratory - ICA *May 2022 – present*

- Provided expertise in the design and construction of semantic search systems based on knowledge graphs, tables, and raw text, using **Python**.
- Advised on the architecture of data provisioning systems, utilizing **REST APIs** and **gRPC** for anchoring data access.

Senior Data Scientist *Rio de Janeiro, Brazil*
Applied Computational Intelligence Laboratory - ICA *March 2018 – April 2022*

- **Led a large-scale R&D project** with a 20-person team over two years, delivering innovative AI solutions that enhanced system trustworthiness and generated measurable impact.
- **Developed transformer-based AI models**, scaling training to millions of simulations across 100 GPUs, using **Torch** and **Tensorflow**.

Researcher *Rio de Janeiro, Brazil*
Applied Computational Intelligence Laboratory - ICA *March 2012 – March 2018*

- Participated in four R&D projects, developing neural networks for client non-compliance classification and optimization algorithms using **C#**, **C++**, and **Bash** on **Linux** systems.
- Configured and optimized compute clusters using PBS Torque and SLURM for efficient resource management.

Consulting Experience

Consultant - Natural Language Processing (NLP) *Lima, Peru*
International Labour Organization (ILO) *March 2021 – September 2021*

- Implemented **NLP solutions** to extract and classify corporate documents, improving data analysis efficiency with **Python**.
- Built an **Information Extraction System** using machine learning to automate the organization of news data.

Consultant - AI Systems *London, UK*
Holistic AI *May 2020 – December 2020*

- Developed **synthetic data generation systems** with **deep learning**, enhancing model training using **TensorFlow**.

Teaching Experience

Assistant Professor - NLP and Data Science *Lima, Peru*
Pontifical Catholic University of Peru (PUCP) *June 2022 – December 2022*

- Taught **Natural Language Processing** using **Python**, focusing on real-world applications in public services and social sciences.

Assistant Professor - Data Science and AI *Rio de Janeiro, Brazil*
Pontifical Catholic University of Rio de Janeiro (PUC-Rio) *March 2016 – April 2022*

- Taught **Decision Support Systems** and **Deep Learning** with **Python**, focusing on practical applications in data science.

Projects

Holistic AI Open Source Library *holisticai - github*

- Spearheaded the development of an open-source toolkit focused on assessing bias, fairness, and explainability in AI systems.
- Led cross-functional collaboration to design and manage GitFlow pipelines and CI/CD workflows, ensuring seamless integration and automation.

Patents

Method for extracting and structuring information *BR 10 2021 023977-8*
Patent number BR 10 2021 023977-8

Publications

Navya Jain, Zekun Wu, **Cristian E. M. Villalobos**, Airlie Hilliard, Adriano Koshiyama, Emre Kazim, Philip Treleaven. From Text to Emoji: How PEFT-Driven Personality Manipulation Unleashes the Emoji Potential in LLMs, Sep 2024.
NeurIPS 2024 Workshop on Behavioral Machine Learning

Mengfei Liang, Archish Arun, Zekun Wu, **Cristian E. M. Villalobos**, Jonathan Lutch, Emre Kazim, Adriano Koshiyama, Philip Treleaven. THaMES: An End-to-End Tool for Hallucination Mitigation and Evaluation in Large Language Models, Sep 2024.
Workshop on Socially Responsible Language Modelling Research

Airlie Hilliard, **Cristian E. M. Villalobos**, Zekun Wu, and Adriano S. Koshiyama, Eliciting Big Five Personality Traits in Large Language Models: A Textual Analysis with Classifier-Driven Approach, Feb 2024.
10.48550/arXiv.2402.08341

Cristian E. M. Villalobos, Kleyton da Costa, Bernardo Modenesi, and Adriano S. Koshiyama. Evaluating explainability for machine learning predictions using model-agnostic metrics, Jan 2024.
10.48550/arXiv.2302.12094

Cristian E. M. Villalobos, Leonardo Forero, Harold De Mello, Cesar Valencia, and Alvaro Orjuela. Sentimental Analysis on Social Media Comments with Recurring Models and Pretrained Word Embeddings in Brazilian Portuguese, Jun 2023.
10.1145/3582768

Cristian E. M. Villalobos, Sara Zannone, Umar Mohammed, and Adriano S. Koshiyama. Uncovering Bias in Face Generation Models, Feb 2023.
10.48550/arXiv.2302.11562

Jose David Bermudez Castro, Ricardo Rei, Jose E. Ruiz, Pedro Achanccaray Diaz, Smith Arauco Canchumuni, **Cristian Muñoz Villalobos**, Felipe Borges Coelho, Leonardo Forero Mendoza, Marco Aurelio C. Pacheco. A free web service for fast COVID-19 classification of chest X-Ray images, Jul 2022.
10.1007/978-3-031-10522-7_29

Potratz, Júlia, **Cristian E. M. Villalobos**, Smith WA Canchumuni and Marco Aurélio C. Pacheco. Deep learning for mapping rainwater drainage networks using Remote Sensing Data, Dec 2021.
5774/4838

Jose D. Bermudez Castro, Smith WA Canchumuni, **Cristian E. M. Villalobos**, Fábio Corrêa Cordeiro, Antônio Marcelo Azevedo Alexandre, and Marco A. Cavalcanti Pacheco. Improvement Optical Character Recognition for Structured Documents using Generative Adversarial Networks, Dec 2021.
10.1109/ICCSA54496.2021.00046

Fábio Corrêa Cordeiro, **Cristian E. M. Villalobos**. Petrolês - How to Build a Specialized Oil and Gas Corpus in Portuguese. Rio Oil and Gas Expo and Conference, Dec 2020.

Cristian E. M. Villalobos, Leonardo Mendoza and Ricardo Tanscheit. Construction of a Transformer-based model for Relation Extraction, Dec 2019
CBIC2019/CBIC2019-117

Costa Leandro Santos DA, Blank Frances Fischberg Oliveira, Fernando Luiz Cyrino, **Cristian E. M. Villalobos**. Conditional Pricing Model with Heteroscedasticity: Evaluation of Brazilian Funds, Aug 2019.
10.1590/S0034-759020190402

Arthur Silveira, **Cristian E. M. Villalobos**, Leonardo Mendoza. Severe Asthma Exacerbations Prediction Using Neural Networks, May 2019.
10.1007/978-3-030-20257-6_10

Leonardo A. Forero Mendoza, **Cristian E. M. Villalobos**, Manoela Kohler, Evelyn Batista and Marco A. Pacheco. Analysis and Classification of Voice Pathologies using Glottal Signal Parameters with Recurrent Neural Networks and SVM, Jan 2019.
10.5220/0007250700190028

Gabriel Lins Tenorio, **Cristian E. M. Villalobos**, Leonardo A. Forero Mendoza,
Eduardo Costa da Silva and Wouter Caarls. Improving Transfer Learning Performance: an Application in the Classification of Remote Sensing Data, Jan 2019.
10.5220/0007372201740183

Skills

Programming Languages: Python, C++, C# JavaScript

Machine Learning Frameworks: TensorFlow, PyTorch, Jax, Scikit-learn

Cloud Platforms: Google Cloud Platform (GCP), Azure

Containerization and Orchestration: Kubernetes, Docker, Singularity

CI/CD Tools: GitHub Actions, GitFlow

Specializations: AI Ethics, Explainability, Bias Mitigation, Large Language Models (LLMs)

Languages

Spanish (Native)

English (Fluent)

Portuguese (Fluent)